

1. PROJECT OVERVIEW

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models

2. FUNCTIONAL REQUIREMENTS (FR) The specific behaviors and operations the ComicCraft system must execute to be considered successful.

- **FR1 (Dynamic Input Form):** The system must offer a responsive HTML UI to collect narrative descriptions and aesthetic preferences from the user.
- **FR2 (LLM Orchestration):** The backend must successfully ping Google Gemini APIs to output both a structured scene-by-scene outline and the corresponding character dialogue.
- **FR3 (Visual Rendering):** The application must interface with the Hugging Face Stable Diffusion endpoint to dynamically generate and download images based on the LLM's scene descriptions.
- **FR4 (Automated Page Layout):** The backend must include a script utilizing the FPDF library to automatically calculate coordinates and stitch the text and images into a downloadable .pdf file.
- **FR5 (Web Presentation):** The system must use a templating engine (Jinja2) to parse backend variables and display the final comic strip directly in the web browser.
- **FR6 (Error Handling):** The backend must feature local text-to-image fallback logic (via Python PIL) to prevent total system failure in the event of third-party API timeouts.

3. NON-FUNCTIONAL REQUIREMENTS (NFR) The qualitative benchmarks and system constraints.

- **NFR1 (Performance):** The entire sequence (from form submission to PDF readiness) must be highly optimized, aiming for a completion time of under 60 seconds.
- **NFR2 (Usability):** The interface must require absolutely zero graphic design knowledge or advanced prompt engineering skills from the user.

- **NFR3 (Asynchronous Processing):** The backend (FastAPI) must execute non-blocking, asynchronous routing to maintain server stability during lengthy API wait times.
- **NFR4 (Environment Security):** Sensitive API tokens must be strictly isolated from the source code via environment (`.env`) variables.